Natural Language & Recommender Systems

COMP 262 Section 401 • Winter 2026 • Group #4

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Project Overview

PHASE 1

Lexicon-Based Sentiment Analysis

- **Dataset Exploration**
  1.7M Amazon reviews
- **Pre-processing**
  Feature selection & cleaning
- **VADER & TextBlob**
  Lexicon-based models
- **Evaluation**
  Confusion matrices & F1
- **Fine Tuning**
  Thresholds & sampling

PHASE 2

ML, Recommender & LLM

- **ML Models**
  Logistic Regression & SVM
- **Cross-Comparison**
  Lexicon vs ML models
- **Recommender System**
  Sentiment-enhanced CF
- **LLM Summarization**
  DistilBART review summaries
- **CS Response Gen**
  Flan-T5 auto-responses

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Phase 1: Lexicon-Based Sentiment Analysis

Amazon Industrial & Scientific Reviews • VADER • TextBlob

Dataset Exploration

1.76M
Total Reviews

165,764
Products

1.25M
Unique Users

4.29
Avg Rating

⭐ Rating Distribution (Heavily Imbalanced)

5 stars	67.9%
4 stars	13.7%
3 stars	6.3%
2 stars	4.1%
1 star	8.0%

📦 Reviews Per Product

Avg 10.61 reviews/product • Median: 2

78,651 products with only 1 review

Most reviewed item: 14,331 reviews

⚠️ Data Quality

78% users wrote only 1 review

Images missing: 98% • Votes: 88%

377,093 duplicate texts • 87,081 full duplicates

ReviewText & Summary: <1% missing ✓

Dataset Pre-processing

Processing Pipeline

1 Feature Selection
Kept: reviewText, summary, overall, asin, reviewerID

2 Label Mapping
4-5★ → Positive
3★ → Neutral
1-2★ → Negative

3 Outlier Decision
Kept <10% outliers — lexicons handle long text

4 Feature Engineering
Created 'full_text' = summary + reviewText

Model-Specific Pre-processing

VADER — Minimal Preprocessing
✓ Preserved capitalization (intensity signal)
✓ Kept punctuation (!!!) as emphasis
✓ Retained degree modifiers ('very', 'extremely')

TextBlob — Light Cleaning
✓ HTML tag removal
✓ Whitespace normalization
✓ Internal tokenization handled

Lexicon Models: VADER vs TextBlob

V VADER

Valence Aware Dictionary & Sentiment Reasoner

How it works:

- Rule-based analyzer for informal text
- Sentiment lexicon with polarity strengths
- Adjusts for caps, punctuation, modifiers
- Detects negations

Outputs:

- Positive, Neutral, Negative proportions
- Compound score ∈ [-1, +1]

Thresholds:

- compound ≥ 0.05 → Positive
- compound ≤ -0.05 → Negative
- otherwise → Neutral

T TextBlob

Pattern-Based Polarity Analysis

How it works:

- Polarity from underlying lexicon
- Polarity range [-1, +1]
- Subjectivity score (not used here)
- Stable baseline model

Limitations:

- Doesn't handle caps/punctuation emphasis
- More conservative on Neutral

Thresholds:

- polarity > 0.1 → Positive
- polarity < -0.1 → Negative
- otherwise → Neutral

Fine Tuning Strategy

Threshold Tuning

VADER: compound at ±0.05

TextBlob: polarity at ±0.1

Balances sensitivity — avoids forcing weak sentiment into Positive or Negative classes

Stratified Sampling

1,000 reviews for evaluation

Mirrors class distribution

Ensures minority classes are fairly represented in testing

Reproducible & meaningful

Preprocessing Per Model

VADER: minimal (keep cues)

TextBlob: light cleaning

Heavy preprocessing avoided — lexicons rely on surface cues that aggressive cleaning removes

ℹ️ No hyperparameter search or model fitting — lexicon models have no trainable weights. Fine tuning is limited to thresholds, sampling methods, and preprocessing choices.

Ground Truth Label Mapping

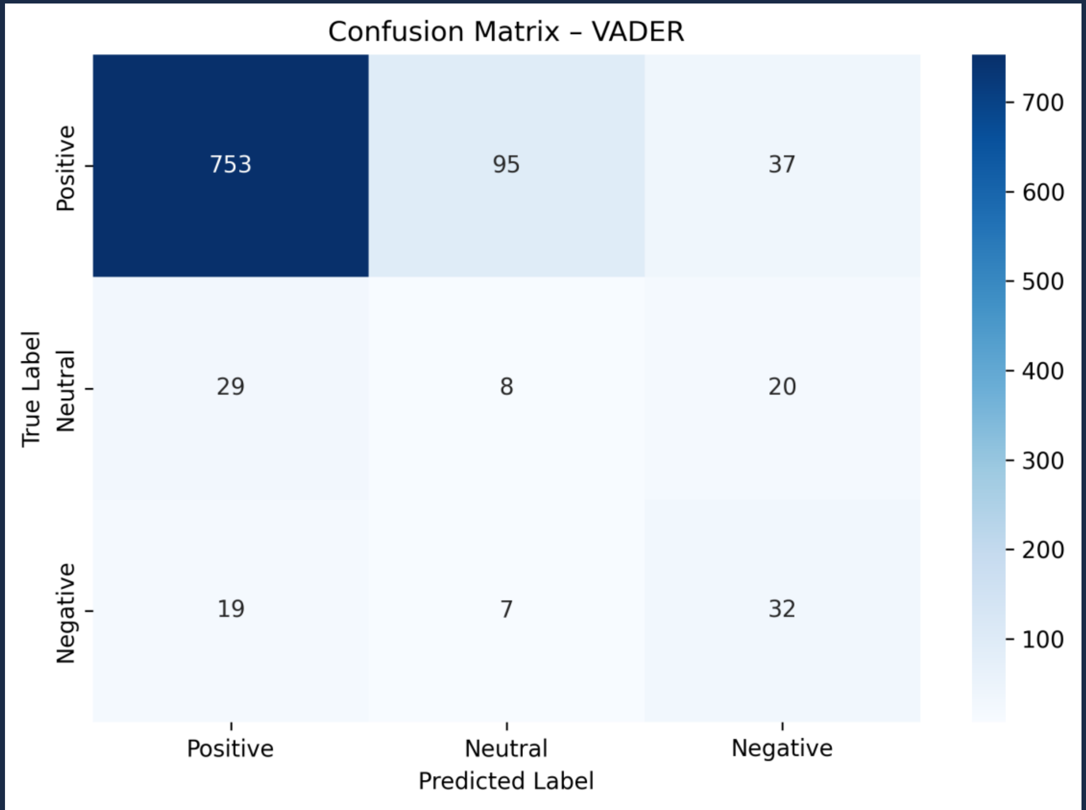
★★★★★ ★★★★ → Positive

★★★ → Neutral

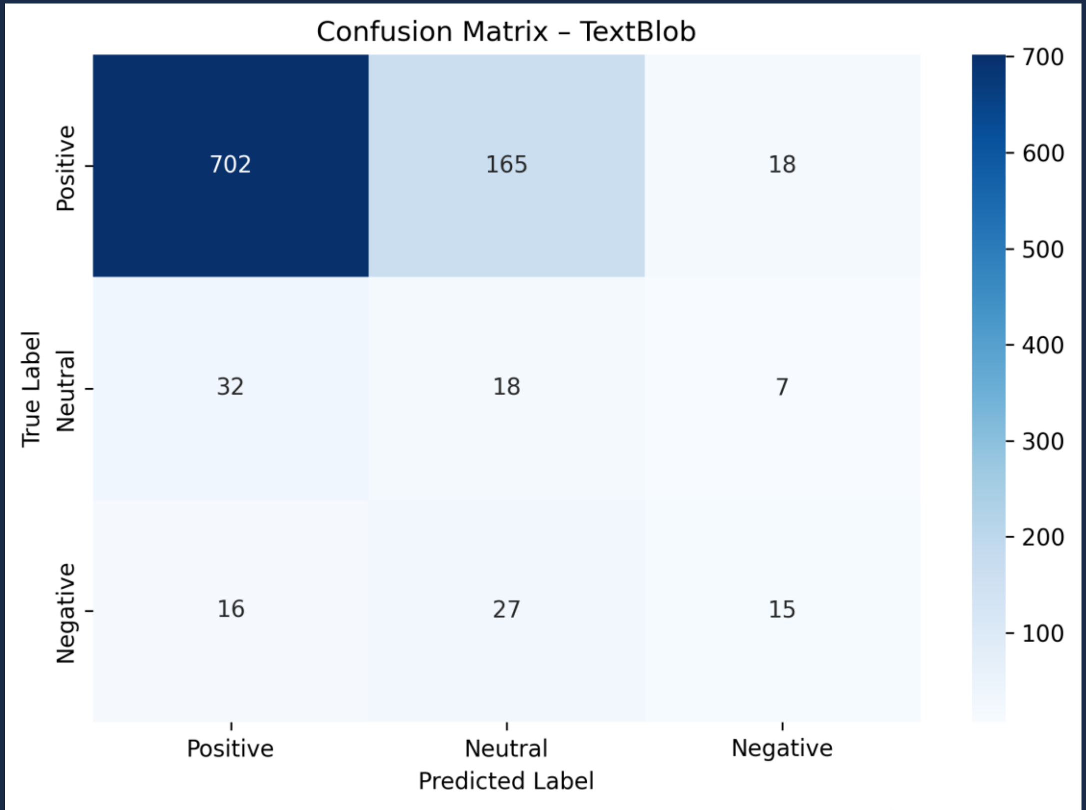
★★ ★ → Negative

Confusion Matrices

VADER



TextBlob



💡 VADER is stronger on the Positive class; both struggle with Neutral near threshold boundaries

Performance Comparison & Distributions

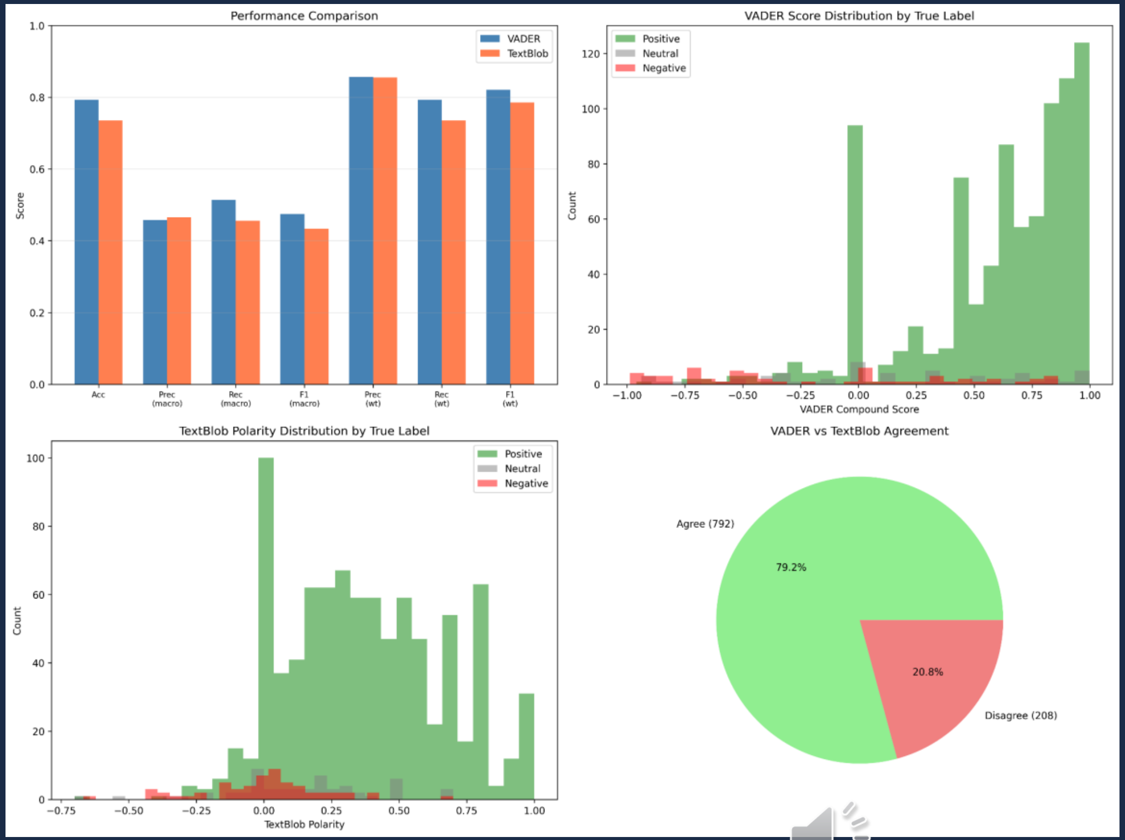
Metrics Comparison

	A	B	C	D
1	Metric	VADER	TextBlob	Difference (V-T)
2	Accuracy	0.793	0.735	0.058
3	Precision (macro)	0.45745091363068885	0.4655714285714286	-0.008120515
4	Recall (macro)	0.5143074909170452	0.455876834	0.058430656809567216
5	F1-Score (macro)	0.4748069890190327	0.4332231686613415	0.041583820357691204
6	Precision (weighted)	0.8569656792645557	0.8549957142857143	0.001969965
7	Recall (weighted)	0.793	0.735	0.058
8	F1-Score (weighted)	0.8212287927594589	0.7854037980515369	0.035824995
9				

Key Findings

- ✓ VADER outperforms TextBlob on Macro F1
- ✓ Macro F1 = primary metric (treats each class equally)
- ✓ ~80% agreement between models
- ✓ Differences mainly on Neutral boundaries

Score Distributions & Agreement



Error Analysis & Phase 1 Conclusion

⚠️ Common Misclassification Patterns

TextBlob: 26.5% misclassified

VADER: 20.7% misclassified

🎭 **Sarcasm / Irony**
"great... not"

⚖️ **Mixed Sentiment**
"good product but terrible shipping"

❓ **Short / Vague**
"ok", "fine"

🔄 **Weak Signals**
Neutral with weak positive/negative words

✅ Phase 1 Conclusions

VADER wins on Macro F1

Better handling of minority classes
Stronger at informal cues & emphasis

Both have limitations

Struggle with ambiguous/neutral reviews
Cannot learn domain-specific sentiment

Next: Supervised ML

Phase 2 improves results with labeled training data & ML classifiers

02

Phase 2: Machine Learning, Recommender & LLM

Logistic Regression • SVM • Collaborative Filtering • Hugging Face

Phase 2: Dataset & Text Representation

Dataset Subset

2,000+ reviews from Industrial & Scientific

Same imbalanced distribution as Phase 1

Predominantly 4-5 star reviews

Stratified splits & macro metrics essential

ML Pre-processing Steps

Lowercase → Remove punctuation → Remove HTML → Tokenize → TF-IDF

+ Combined summary & reviewText

TF-IDF: Text Representation Method

Sparse Text

Performs well on high-dimensional text

Domain Terms

Captures vocabulary like 'defect', 'broke'

Linear Models

Efficient with Logistic Regression & SVM

Scalable

Computationally efficient for large datasets

Machine Learning Models & Results

Logistic Regression

Linear classifier that models class probability boundaries

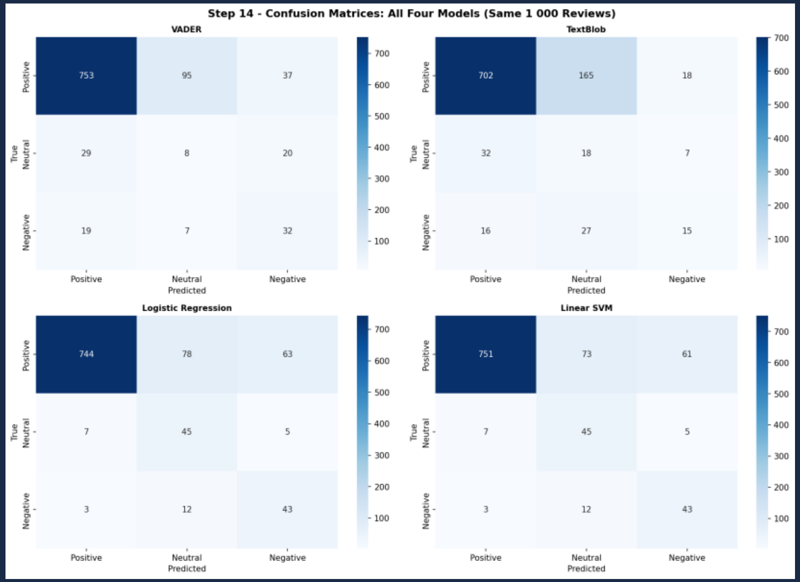
Linear SVM

Maximum margin classifier for high-dimensional text

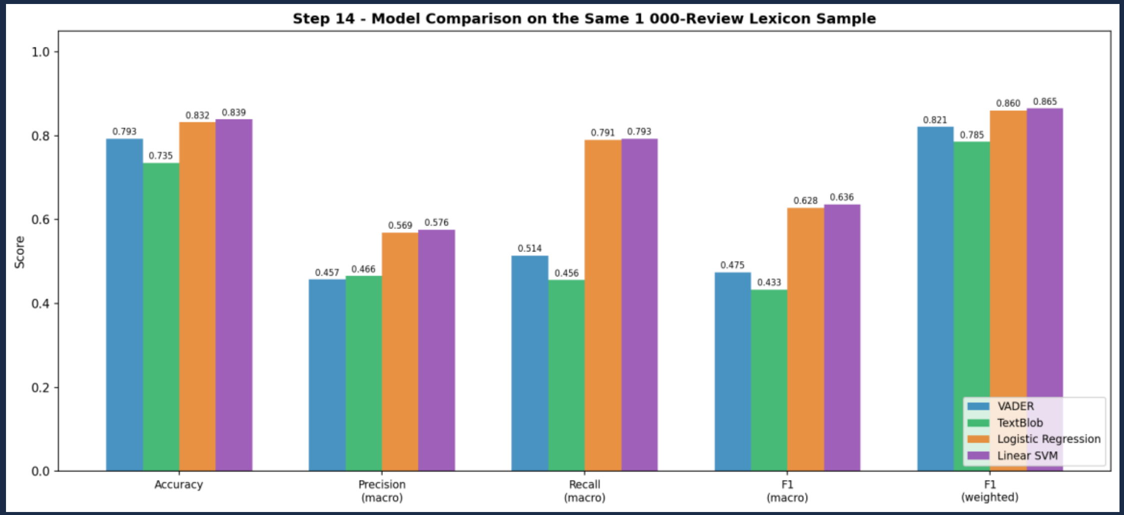
Training Configuration

70/30 stratified train/test split • Grid Search tuning • Limited TF-IDF vocab size

Results



Comparison



Cross-Model Comparison: All 4 Models

All models tested on same 1,000 stratified sample for fair comparison

ML models outperform lexicon models across all metrics

Lexicons struggle with mixed sentiment & domain terms • ML learns patterns like 'broke', 'defect', 'return'

1 Linear SVM

Best Macro F1 overall

Ideal production model

2 Logistic Regression

Close second

Strong across all metrics

3 VADER

Best lexicon model

Good at informal cues

4 TextBlob

Conservative baseline

Over-predicts Neutral

Sentiment-Enhanced Collaborative Filtering

Enhancement Architecture

Review Text → VADER Analysis → Sentiment Score → Blend Formula → Enhanced Rating

Formulas

```
sentiment_score     = (VADER_compound + 1) / 2
normalized_star     = (rating − 1) / 4
enhanced_rating     = 0.6 × norm_star + 0.4 × sent_score
```

Results

MAE Improvement

▼ 6.7%

RMSE Change

▲ 24.9%

Sentiment minimizes noise in sparse rating matrices → more accurate & stable recommendations

Especially helpful for controversial products

03

LLM-Powered NLP Tasks

Review Summarization • Customer Service Response Generation

Review Summarization with LLM

🤗 **sshleifer/distilbart-cnn-12-6**

Pipeline:

Select 10 reviews >100 words → **Feed to DistilBART model** → **Generate abstractive summary** → **~50 word condensed output**

Example 1

Original (~120 words):

"As another reviewer mentioned I don't think there is any difference in which Goop you buy... The only reason I dropped off a star is that it often hardens after a tube is opened..."

Generated Summary (38 words):

"The only reason I dropped off a star is that it often hardens after a tube is opened even though the cap is on tight. Other than that, the product itself is 5-stars all the way."

Observations

- ✓ Captures main sentiment & key aspects
- ✓ Coherent, preserves essential meaning
- ✓ Significantly reduces text length
- △ Minor details sometimes omitted
- △ Occasionally shorter than 50-word target
- △ Focuses on dominant sentiment

💡 Abstractive summarization generates new sentences — more natural & readable than extractive methods

Automatic Customer Service Response

🤗 **google/flan-t5-large (instruction-tuned)**

Customer Review → **Extract Complaint** → **Design Prompt** → **Generate Response**

Sample Result

📩 **Customer Review:**
"I've found all kinds of uses for this product... however when removing it, the adhesive pulled paint off the wall..."

🔍 **Extracted Complaint:**
"The adhesive pulled paint off walls and is not wall-friendly like other products."

💬 **Generated Response:**
"Dear Customer, thank you for taking the time to share your experience... We sincerely apologize for the issue you described... Please contact our support team and we will assist you further."

✓ Coherent • Polite • Well-structured • Acknowledges concerns

Strengths

△ Generic responses • No business context • Not always actionable

Limitations

Conclusions & Future Work

1 Phase 1: Lexicon

✓ VADER outperforms TextBlob
(higher Macro F1)
✓ Good baseline without training
△ Struggles with Neutral reviews
△ Cannot learn domain sentiment

2 Phase 2: ML

✓ Linear SVM = best model
✓ ML beats lexicon on all metrics
✓ Sentiment CF improves MAE 6.7%
△ Neutral class remains hardest

3 LLM Tasks

✓ Effective summarization
✓ Coherent CS responses
✓ Extends beyond classification
△ Pre-trained, not fine-tuned
△ Can lack domain specificity

🚀 Future Improvements

🔬 Fine-tune LLMs on domain-specific datasets

🔗 Integrate retrieval-based systems for precise responses

🧠 Explore deep learning (BERT, transformers) for classification

⚡ Strong potential for real-world integration of all approaches

Thank You!

Questions & Discussion

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